

Self-Paced Brain-Computer Interface on Sensorimotor Rhythms for Controlling Virtual Objects

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Abstract—Non-invasive electroencephalogram (EEG) based Brain-Computer Interface (BCI) systems have become a fascinating study area in many research fields. A majority of the research in this area is conducted synchronously. Hence, at the time of the experiment, the user's state of mind tends to be different from its natural behavior. As a solution to this problem, self-paced BCI systems started gaining popularity in recent years. However, certain challenges remain to be addressed even with this method. Most of the research on self-paced BCI systems is focused on motor-imagery control, whereas research on non-motor imagery mental tasks is limited. However, individuals with severe paralysis may face challenges in performing motor imagery tasks. In this research, we explore the possibility of using the techniques used in the motor-imagery method for non-motor imagery mental tasks. The intention here is to use them in virtual object-controlling applications. The research was done with five different classification models using features from the Fast Fourier Transform (FFT) and Wavelet Transform (WT). The K-nearest neighbor model with features obtained with FFT continuously sustained its performance with a high 55% true positive rate.

Keywords—brain-computer interface, self-paced, EEG, sensorimotor rhythms, machine learning

I. INTRODUCTION

Brain-Computer Interfaces (BCI) have come into the limelight in recent years as a source of innovative applications in many fields, from medicine to the gaming and robotics industry. New applications for BCIs are continually emerging, with one notable direction being the use of BCIs to control virtual

environments. These virtual environments offer researchers an efficient platform for developing and testing BCI applications, in contrast to the early days when custom hardware had to be built for these purposes.

Identifying and classifying the thought commands of a person is a significant task of a BCI system. Laboratory-based BCI systems often use a synchronous method where the subjects are guided toward certain intentions by visual or verbal cues. Not only is this method far from the natural behavior of the mind, but it can also have certain limitations when it comes to practical application. It relies on predefined time and task indicators and is less user-friendly due to the requirement for interactions to occur within a specific timeframe in response to cues from the system. Alternatively, researchers have introduced self-paced systems where the subjects can decide when to have controlling commands [1]. Here, the BCI recognizes them based on the state of their thought commands, which may belong to either an intentional-control (IC) state or a no-control (NC) state. However, self-paced BCI systems are more challenging than the former as there are major concerns with the false positive rate (FPR) of signal classification.

The most common approach used in research for BCI is non-invasive EEG-based devices [2], [3]. The key advantages of non-invasive techniques are the maturity of the technology, relative ease of use and low cost, and the robustness, portability, and versatility of recent EEG devices [4]. Among the different EEG brain activity patterns used for BCI ap-

proaches, there are P300 [5]–[7], steady-state visual evoked potentials (SSVEP) [8], [9] and event-related desynchronization/synchronization (ERD/ERS) [10], [11]. The proposed research focuses on ERD, utilizing the ability to control virtual objects through mind intent instead of relying on external visual or verbal cues.

The motor imagery (MI) and motor execution (ME) methods have been the most used asynchronous methods in BCI research. In the MI method, the subjects imagine the movement of their arm or leg without moving them physically, whereas in the ME method, they execute a movement of a limb. In 2007, Leeb et al. [12] conducted ERD-based research where MI was used to operate a person in a virtual environment, which resulted in a 50% true positive rate (TPR) and an average FPR of less than 10%. The classification method used here was linear discriminant analysis (LDA). In 2010, they increased TPR to 79% and reduced the FPR to 0.67% per minute [13] with the usage of Support Vector Machines (SVMs). However, there are cases where some subjects, such as patients who have been paralyzed for long periods, are unable to obtain the expected results. Researchers are interested in controlling virtual objects using thoughts of changing the spatial transformation of the object rather than motor imagery or motor execution. For example, thoughts of moving an object into a predefined direction, rotating around an axis, or scaling the object are considered spatial transforms or non-MI mental tasks in this research [14]. This area is still novel to the field of BCI research. Research by Faradji et al. [6] explored the idea of rotation of a virtual object in 3D space using self-paced mind-intent. They used auto-scalar autoregressive methods for feature extraction, and the classification was performed using quadratic discriminant analysis. They obtained TPR values within the range of 27.36% - 64.73% and 0.01% FPR. Although there are numerous types of research on using MI to control virtual objects that give us higher accuracy [14], research done by Faradji et al. [6] explores the possibility of controlling objects more naturally. Therefore, although the TPR is relatively low compared to MI-related research, this method is preferable in real-time applications.

In this research, we explore the potential of applying signal acquisition, feature extraction, and classification techniques from MI studies to non-MI mental tasks, aiming to use these signals for controlling virtual objects through a self-paced BCI system. We tested different artifact removal methods, such as Independent Component Analysis (ICA), filtration methods, and wavelet transformation with a threshold. Feature extraction was performed using signal representations such as the Fast Fourier Transform (FFT) and Wavelet Transform (WT) of EEG data. Their ability to perform in real-time and their information resolution were obtained using different methods. We used Random Forest, Catboost, K-Nearest Neighbour Algorithm (KNN), Support Vector Machine (SVM), and Quadratic Discriminant Analysis (QDA) for classification. According to the results obtained in this research, we present a comparison between the results of different approaches that were tested out.

II. METHODOLOGY AND EXPERIMENTS

In the methodology section, we first discuss the experimental protocol that was followed. Then, we look at different signal representations implemented for feature extraction purposes. Furthermore, this section includes the artifact removal mechanisms of EEG data and the classification methods implemented.

A. Experimental Setup

Given the novelty of this research area, we chose a single subject to explore the potential of using motor-imagery techniques for non-motor mental tasks, focusing on mind-controlled virtual object manipulation. Our subject was a male volunteer, age 24. First, we trained the subject to train three mind intents, left, right, and none, with a virtual stationary object (sphere). This training was done in a limited time trial, such as 10 seconds, because the performance of the mental task degrades over time. We used an OpenBCI Cyton board with 8 EEG channels to capture the subject's brain waves. This setup is shown in Fig. 1. EEG signals were fed for processing and denoising. We used the OpenBCI GUI to send EEG time series signals into a Python application through the Lab Streaming Layer (LSL), where we did the post-processing and classification of data. Signal processing and classification are done in a computer with an Intel Core i5-6200U processor.

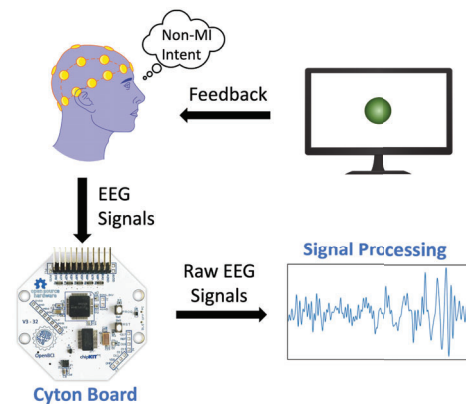


Fig. 1. Experimental setup to acquire EEG signals

B. Electrodes and Electrode Placement

We used eight Golden cup electrodes to sample EEG data. We placed those on the subject according to the 10–20 method [15]. The 10–20 system is an internationally recognized method to describe and apply the location of scalp electrodes in the context of an EEG exam. The brain waves related to controlling virtual objects are induced in the motor cortex, so electrode placement positions are chosen to extract the maximum amount of information. In our experiment, we placed electrodes in the positions indicated by the red circles in Fig. 2.

A1 and A2 in Fig. 2 act as the reference voltage (ground), which will be the references for all the EEG electrodes placed on the head. These were placed on the earlobe as they are

considered a neutral electrical reference compared to other parts of the body. Nasion and inion were used as landmarks for the positioning of EEG electrodes.

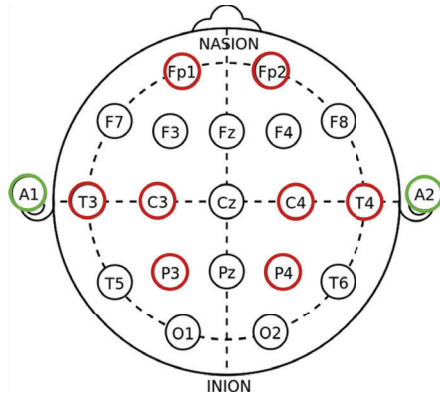


Fig. 2. 10-20 Experimental electrode placement

C. Virtual Environment

Virtual objects meant to be controlled were created with the Unity game engine. The subject is trained in a virtual environment where the display is a 15.6-inch monitor with a resolution of 1920 x 1080 and a refresh rate of 60Hz. Data of mind intent will be recorded where the subject will focus on moving the objects along horizontal axes. Fig. 3 shows the virtual environment that was created for the experiment.

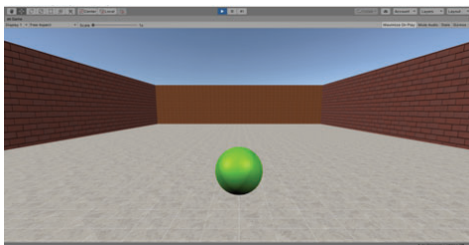


Fig. 3. Virtual environment

D. Signal Acquisition and Pre-processing

Data were collected with 10-second trials separated by 30-second intervals. Initially, the EEG time series signal is notched at 50Hz using the OpenBCI GUI to filter out the AC current frequency before being transferred to the Python application. This data was saved in .csv format. Data manipulation was done according to the different artifact removal methods that were followed. Unfiltered time series data was used for ICA and post-artifact removal signal representations. In this research, our area of interest is 15-60Hz for EEG signals. This range is chosen because the beta (13-30Hz) and gamma (30Hz and above) signal ranges are the important signal ranges in non-MI EEG signals [16]. The signals above 60 Hz are contaminated with other artifacts and do not contain interesting data, so information that can be extracted from them is low and noisy [17].

E. Signal Representations

We used two main signal representation methods in this research. They are the Fast Fourier Transform (FFT) and Wavelet Transform (WT). Different research in the BCI domain has used different representations of data for feature extraction [17]. Some are FFT, WT, time series data, statistical data, and surface Laplacian. Out of them, we chose FFT and WT since our focus is on exploring the behavior of EEG signals in the frequency domain and how they behave when combined with the time domain.

- 1) Fast Fourier Transform: FFT is a signal representation that contains frequency spectral details of a time series signal. It includes the magnitudes related to each frequency component of the signal. Given that the FFT function returns a symmetric spectrum for real signals, we extracted features from the positive frequencies. Since our research focuses on the 15-60 Hz frequency band, we extracted the FFT for this range, with a sample shown in Fig. 4.

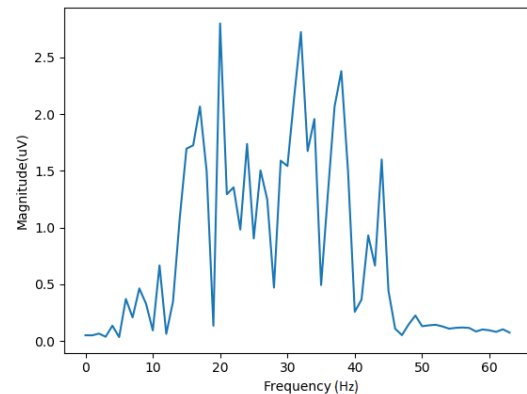


Fig. 4. Extracted FFT representation 15 - 60 Hz

- 2) Wavelet: Daubechies 4 (db4) wavelet is used as the mother wavelet in this research due to its suitability for processing biomedical signals [18]. The input signal has a frequency band of 0-250 Hz. Since the relevant frequency band lies in the Gamma and Beta (16-63Hz), our area of interest is 15-60Hz for EEG signals. Therefore, the filtered signal will be decomposed only up to 5 levels to obtain the detailed coefficients related to the beta and gamma bands. CD4 and CD3 are the detailed coefficients representing Gamma and beta bandwidth relevant to non-MI. These coefficients can be used as features for the subsequent classification method. This method can also be used to extract required frequency bands by reconstructing the signals after removing unwanted detailed coefficients and artifact ranges.

F. Artifact Removal

The ocular artifact was a primary focus of this research. To address this issue, we employed both the ICA and the Wavelet Transform Threshold Method. Specifically, we used

the ICA algorithm to remove the Electrooculography (EOG) artifact. This involved applying ICA to a time series signal that had been arranged into a 2D array (channel count x time series signal length). The application of ICA yielded separate independent components, which are plotted in Fig. 5.

In Fig. 5, the signal intensity of the independent component is positive in the red area and negative in the blue area. Each brain area shown in this figure represents a separate independent component. We identified ICA001 as the independent component that contains the EOG artifact. We used the ICA algorithm to remove the EOG artifact, as shown in Fig. 6. To acquire the real signal representation.

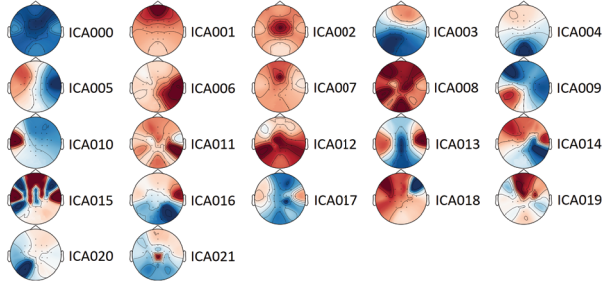


Fig. 5. Independent components mapped to the brain area

We used the Wavelet Transform Threshold Method to remove high-frequency noises. However, artifacts of EEG signals cannot be removed using the universal threshold method directly. FFT and Wavelet transform can also be used as artifact removal methods (after post-signal representation). FFT can be used to reconstruct the signal after removing unwanted frequency coefficients related to artifacts. WT can be used to reconstruct the signal after removing unwanted detailed coefficients related to artifacts.

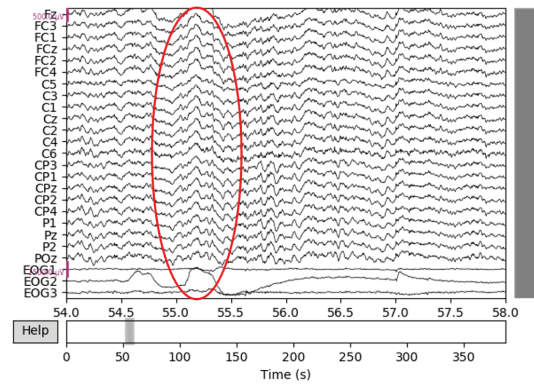
G. Feature Extraction

Frequency bins from 15 Hz to 60 Hz (i.e., 15 Hz, 16 Hz, 17 Hz, ...) are used as features extracted from FFT for training classification models. Detailed coefficients on the 4th and 3rd levels are used as features extracted from the wavelet transform method.

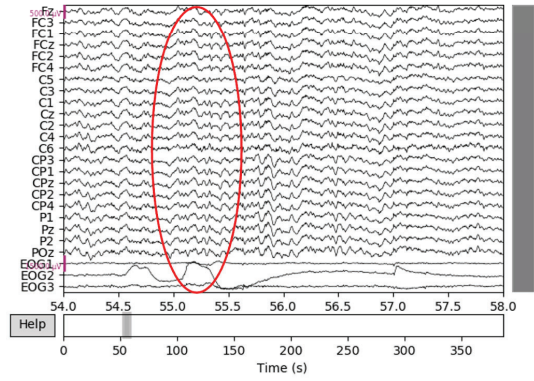
H. Classification

After performing model-based feature selection methods, the extracted features were applied to the models. We used Random Forest Classifier, Catboost, KNN, SVM, and QDA models for classification. The classification is based on three classes: Left, Right, and None. The evaluation of classification models is done by cross-validation. We used 4800 data points to train the model, and for the test set, we used 480 data points. To select the test set, we used randomly picked data points from 5 days, selecting 96 data points from each day. The following were used as evaluation metrics.

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}} \quad (1)$$



(a)



(b)

Fig. 6. EEG signals with EOG artifacts (a) and without EOG artifact (b)

$$\text{True Positive Rate} = \frac{\text{True Positives}}{(\text{True Positives} + \text{False Negatives})} \quad (2)$$

III. RESULTS

A. Real-time Requirement Results

Table I compares the different artifact removal and feature extraction methods on computational cost and time taken to calculate 25 instances of 8-channel EEG data. The results show that the FFT frequency coefficient removal method has the lowest computational time, so it is preferable for real-time usage.

B. Classification Results

Frequency bin components extracted by FFT and detailed coefficients extracted by WT were used as features for classification purposes. In Table II, we have compared the accuracies between different classification models. The best hyperparameter combination for each model is determined by a grid search using 10-fold cross-validation as an evaluation method. The KNN model, which had features obtained with FFT, showed the highest accuracy. As shown in Table II, the overall accuracy obtained when using FFT is higher than when using WT. We then received the accuracy with respect to each mind intent obtained with FFT. Table III gives the TPR of

TABLE I
COMPARISON OF FEATURE EXTRACTION AND ARTIFACT REMOVAL
METHODS

Methods	Computational Cost k – no. of sources n – no. of samples	Computational Time /ms
ICA	$O(kn \log n)$	86
FFT frequency coefficient removal method	$O(n^2)$	23
WT threshold method (universal / Bayesian threshold)	$O(n^2 \log n)$	53
WT detailed coefficient removal method	$O(n^2 \log n)$	42

each class obtained with FFT with respect to each model. The confusion matrix of the KNN model is shown in Fig. 7. Since we have data collected over five days, we used a 5-fold cross-validation to estimate the consistency of accuracies. This is shown in Table IV.

TABLE II
COMPARISON OF FEATURE EXTRACTION AND ARTIFACT REMOVAL
METHODS

Method	Random Forest	Catboost	KNN	SVM	QDA
FFT	0.4740	0.5000	0.5521	0.4519	0.4688
WT	0.3854	0.3802	0.4063	0.4323	0.4063

TABLE III
TRUE POSITIVE RATES OF EACH CLASS WITH RESPECT TO MODEL

Class	Random Forest -FFT	Catboost -FFT	KNN -FFT	SVM -FFT	QDA -FFT
LEFT	0.79	0.57	0.67	0.68	0.79
RIGHT	0.36	0.47	0.51	0.44	0.49
NONE	0.06	0.44	0.44	0.00	0.06

TABLE IV
FIVE-FOLD CROSS-VALIDATION (CV) WITH FFT AND WT

Metric	Random Forest	Catboost	KNN	SVM	QDA
FFT CV_score	0.55	0.54	0.56	0.54	0.55
FFT std_dev	0.07	0.01	0.02	0.07	0.09
WT CV_score	0.47	0.50	0.47	0.49	0.47
WT std_dev	0.09	0.05	0.03	0.09	0.06

IV. DISCUSSION

A. Real-time Requirements Analysis

Since EEG signals are in the micro voltage range, they are vulnerable to artifacts. As this is a real-time application,

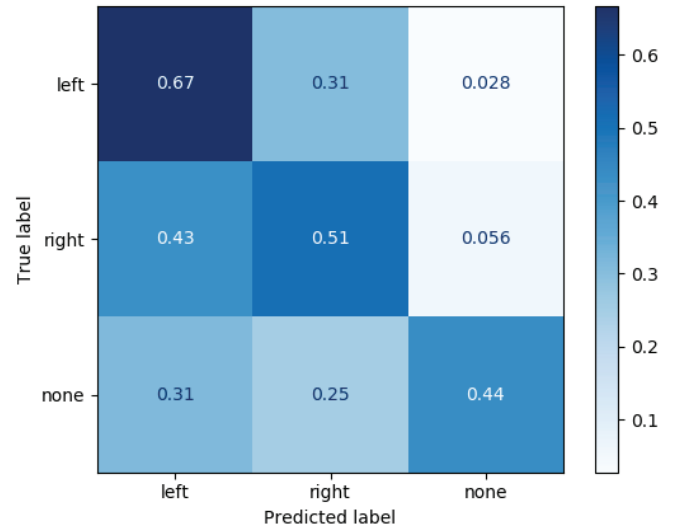


Fig. 7. Confusion matrix of KNN model

we were more concerned about the time complexities and processing costs of each signal analysis and processing method in this research. ICA can be considered one of the best methods for offline applications since it is more sensitive to discovering artifacts and has the capability to preserve more information while removing artifacts. However, the ICA method had a higher computational time than other methods. Plus, to identify the signals that are related to the artifacts, it is a must that training data is contaminated with those artifacts. The WT threshold method has a lower artifact removal ability than other methods. In our research, we tested out the universal threshold method. Since the universal threshold is sensitive to higher frequencies, it is not applicable for eliminating lower-frequency components. Due to the limitations of these two methods, we have chosen the WT detailed coefficient removal method and the FFT frequency component removal method as our artifact removal methods.

B. Feature Extraction and Classification Analysis

Overall, when it comes to signal classification, FFT frequency components as features gave higher accuracy than WT detailed coefficients as features. Although Catboost and KNN with FFT show significantly higher accuracy than other models, all five classification models show high TPR when identifying categorical values left and right. However, the TPR of non-class (non-intended state) identification is low on QDA, SVM, and Random Forest algorithms. KNN and Catboost models show a significant TPR in identifying non-intended states. When comparing both KNN and Catboost models, KNN has overall higher accuracy and TPR than other models. Another observation we made is that the Random Forest algorithm requires more memory; it also requires considerable time to predict instances. For one instance, it takes approximately 110 ms. Therefore, it is not an ideal approach for real-time classification. According to the results obtained by 5-fold

cross-validation, the Catboost algorithm and KNN sustain their performance more continuously than other models.

Comparing our results directly to those in the literature is not fair due to differences in datasets, number of subjects, and task types. For instance, Faradji et al. [6] achieved a 54.6% accuracy using auto-scalar auto-regressive methods and quadratic discriminant analysis with 29 EEG channels, averaging across multiple tasks. In contrast, our study achieved 55% accuracy with the KNN algorithm and FFT feature extraction using only 8 EEG channels.

C. Limitations and Future Directions

This research primarily explores the potential of applying MI techniques to non-MI mental tasks using a single subject. Due to the focus on a single individual, the findings may not be broadly generalizable. To enhance the generalizability of the classification, future research should involve a larger and more diverse participant pool to examine subject-independent performance in non-MI mental tasks.

Additionally, while this research utilized machine learning techniques to analyze MI methods in non-MI tasks, there is significant potential for improvement by incorporating advanced deep learning techniques. Future studies should investigate how deep learning algorithms can enhance feature extraction and classification performance, potentially leading to more accurate and efficient BCI systems.

V. CONCLUSION

In our research, we explored advancements in MI-based research and assessed their applicability to self-paced non-MI-based applications. Our investigation revealed that while new signal analysis methods have enhanced the detection of thought commands, some fall short in their application to real-time, self-paced BCIs.

We assessed the effectiveness of five classification models—Random Forest Classifier, Catboost, KNN, SVM, and QDA—in categorizing signals into three distinct classes: Left, Right, and None. The feature extraction process utilized signal representations of EEG data through FFT and WT. Compared to FFT features, WT features exhibit lower reliability in non-MI signals, highlighting WT's limited effectiveness in identifying thought commands in self-paced BCIs.

Among classification models, KNN with FFT extraction is ideal, yielding a 55% TPR. Catboost and KNN show consistent performance compared to prior research, with less standard deviation. Hence, machine learning methods show promise in MI-based research and could be explored for non-MI intent in self-paced BCIs. Combining WT and FFT features is another avenue for further exploration in self-paced BCI performance enhancement.

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